

■ Smart Tour Recommender System - Project Proposal

1. Executive Summary

The **Smart Tour Recommender System** is an advanced, data-driven platform designed to revolutionize tourism in Pakistan. By leveraging Hybrid Machine Learning (Content-based + Collaborative Filtering), the system provides personalized travel suggestions based on user preferences, budget constraints, and safety profiles. Featuring a modern "Glassmorphism" UI, interactive geospatial mapping, and deep analytics, it offers a seamless and immersive experience for travelers exploring the country's diverse landscapes.

2. Core Features

■ Smart Recommendation Engine

- Hybrid Logic using TF-IDF vectorization and Collaborative Filtering.
- Dynamic filtering by budget, province, and safety rating.
- Weighted scoring based on ratings, popularity, and experience.

■ Immersive User Interface (UI/UX)

- Glassmorphism design with Gaussian blur effects.
- HD video backgrounds of Pakistani landscapes.
- Responsive sidebar navigation.

■ Interactive Geospatial Mapping

- Pydeck-based interactive 2D/3D maps.
- Adaptive dark/light themes.
- Auto zoom and focus on selected destinations.

■ Advanced Analytics Dashboard

- Pie charts, bar charts, and scatter plots using Plotly.
- Real-time metrics and safety analysis.

■ Deep Dive Exploration

- Province-wise destination browsing.
- Detailed insights with images and categorical tags.

3. Technology Stack

Layer	Technology	Purpose
Frontend Framework	Streamlit	Rapid development of data-driven UI
UI Styling	Custom CSS & HTML	Glassmorphism and video backgrounds
Data Processing	Pandas & NumPy	Data cleaning and aggregation
Machine Learning	Scikit-learn	TF-IDF, Cosine Similarity, NMF
Data Visualization	Plotly & Seaborn	Interactive charts and plots
Geospatial Mapping	Pydeck	Spatial rendering and mapping
Backend Language	Python 3.10+	Core application logic

4. Project Structure (File Mapping)

Core Files

- **app.py**: Main application entry point.
- **raw1.py**: Backend engine with ML and analytics logic.
- **requirements.txt**: Dependency manifest.

Directories

- **/data/**: CSV datasets for destinations, users, and reviews.
- **/assets/**: Images and travel video assets.
- **/outputs/**: Reports and generated charts.

5. How It Works (Workflow & Logic)

1. Data Ingestion & Processing

DataLoader and DataProcessor clean and merge datasets.

2. Feature Engineering

Destination tags are converted into numerical vectors using TF-IDF.

3. Recommendation Logic

User preferences are transformed into vectors and matched using Cosine Similarity. Safety and budget filters are applied before ranking.

4. Interactive Rendering

Results are displayed as Glass Cards with map visualization using Pydeck.

5. Analytics Generation

Plotly generates charts and statistics for insights.

6. Installation & Deployment

Install Dependencies:

```
pip install -r requirements.txt
```

Launch Application:

```
streamlit run app.py
```

Access:

Open `http://localhost:8501` in your browser.

7. Future Roadmap

- Real-time weather integration.
- Route optimization using Dijkstra's algorithm.
- Mobile app conversion using Flutter or React Native.